AROB

General introduction

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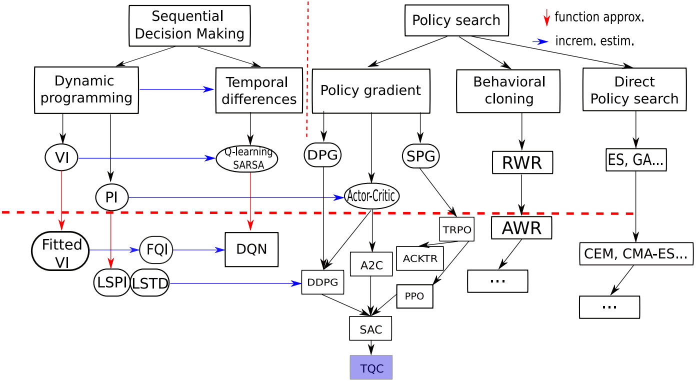
Content

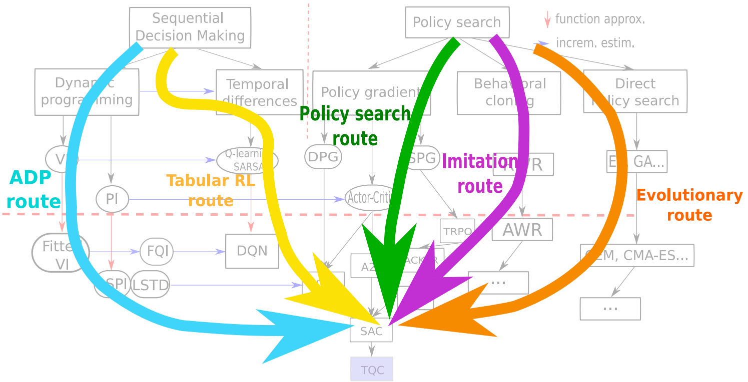
1. Regression
2. Model-free reinforcement learning (tabular)
3. Statistics in RL, hyper-parameter tuning
4. Model-based reinforcement learning (tabular) (+ bio modelling?)
5. Evolution strategies (ES, CEM, CMA-ES)
6. Evolutionary robotics
7. Collective evolutionary robotics
8. Multi-objective evolutionary optimization
9. Evolutionary optimization of structures
10. Diversity seeking methods (novelty search, QD)

The Big Picture



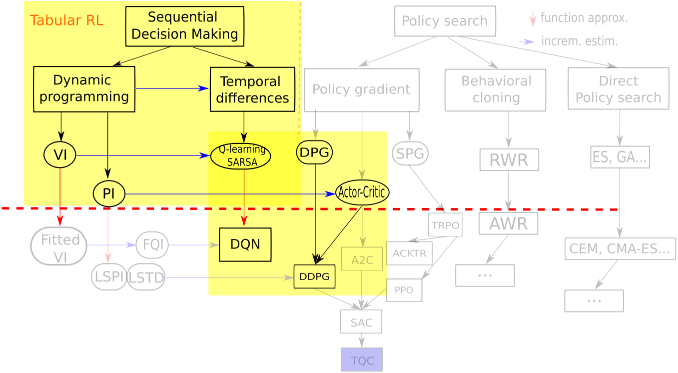
- A very partial view of the whole RL literature

The five routes to deep RL



- Five different ways to come to Deep RL

The Tabular RL route



- The favorite route of beginners
- Start from Sutton&Barto, present Q-learning, SARSA and Actor-Critic
- Add function approximation with NNs, go to DQN, DDPG, TD3 (in M2)

Sutton, R. S. & Barto, A. G. (1998) *Reinforcement Learning: An Introduction*. MIT Press.

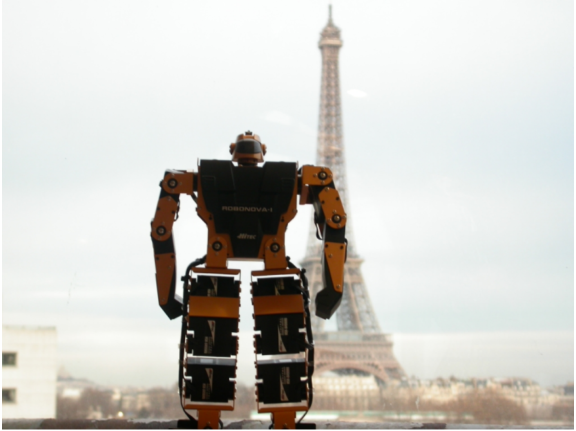
Organization (my part)

- Lessons about algorithms, concepts and tools
- During lessons: rehearsal questions about the concepts
- Labs: coding algorithms, practicing, tuning hyper-parameters (notebooks)
- Reports: studying experimental phenomena
- Lessons are available in video
- Everything available in advance, supplementary material...

Evaluation

- One lab per week, average over 10 labs (small tweek in my part)
- Questions about lessons are not evaluated
- Delay is penalized (-2 point over 20 per day for any deadline missed)
- Technical details on the Moodle page
- Advices:
    - Watch lessons in advance, ask questions in classroom
- Look at the Moodle page for more details

Any question?



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Sutton, R. S. and Barto, A. G. (1998).
Reinforcement Learning: An Introduction.
MIT Press.